

Restoring the Image Does Not Restore Navigation: Diagnosing Degradation-Robust Object-Goal Navigation as a Commitment Problem

Anonymous submission

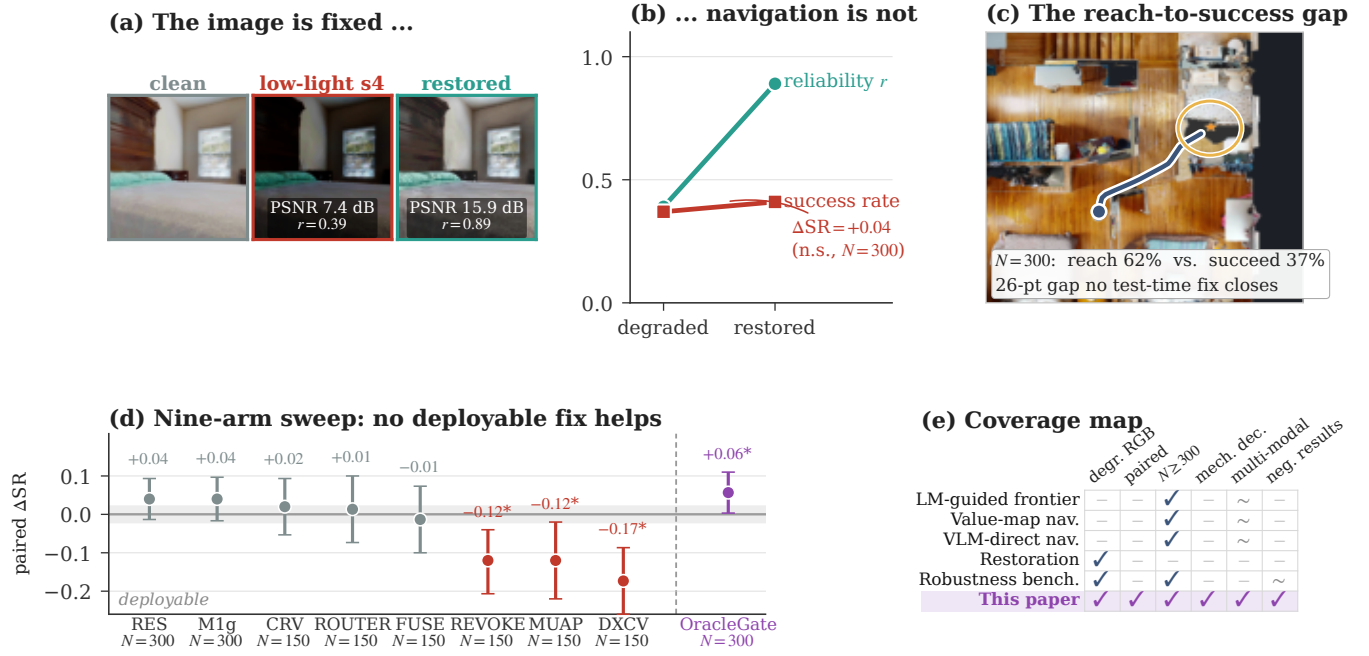


Figure 1: **Restoring the image does not restore navigation.** (a) Blanket restoration recovers image quality (PSNR 7.4→15.9 dB; blind reliability at the committing frame 0.39→0.89). (b) The same restoration moves success by only +0.04, indistinguishable from no intervention ($N=300$ paired, McNemar $p=0.182$). (c) *Background*: one representative HM3D ObjectNav episode under low-light severity 4; bright path traces the real rendered trajectory. *Annotations*: aggregate statistics on the $N=300$ main split—the agent reaches within one metre of the goal in 62% of episodes but succeeds in only 37%, a 26-point reach-to-success gap that no perception-side intervention closes. (d) The full nine-arm intervention sweep (companion to Tab. 1): markers are paired ΔSR vs. the frozen baseline, bars are bootstrap 95% CIs, * marks McNemar $p<0.05$. Every deployable arm is either indistinguishable from zero or *significantly negative*; the only significant positive is the non-deployable per-frame appearance oracle, capped at +0.057.

Abstract

Zero-shot object-goal navigation agents driven by vision-language models lose much of their success rate once the camera input is corrupted by low light, haze, or motion, and the default remedy is to restore the image before it reaches the policy. Through a controlled study on InstructNav in Habitat-Matterport 3D ObjectNav, with a same-seed paired protocol at $N=300$ and a blind per-step reliability estimator, we show that this remedy targets the wrong layer. Degradation does not stop the agent from acting; it corrupts the *quality* of an irreversible commitment: the agent still reaches

within one metre of the goal in 62% of episodes yet succeeds in only 36%, and restoration that visibly repairs the image moves success by a non-significant +0.04. A sweep over six causal families of test-time interventions—pixel restoration with a per-frame appearance upper bound, reliability-gated reweighting, commit-timing revocation, forced commitment, and temporal/multi-view aggregation—finds that every deployable family lands at ~ 0 or *significantly hurts* success, evidence for an irreducible commitment error that no test-time channel recovers. The fingerprint survives a cross-corruption replication (low-light, motion-blur, fog, Gaussian noise) on the

same InstructNav+Qwen2-VL-7B pipeline, and we document that per-condition restoration effects can flip sign between $N=40$ and $N=150$, so under-powered evaluations can conclude the wrong direction. These results point the field away from test-time patches on frozen, clean-trained agents, toward degradation-aware training and learned multi-modal sensing. We release the benchmark, protocol, and all negative results.

Introduction

Object-goal navigation requires an agent to locate an instance of a target category in an unknown environment, perceiving, deciding, and moving in a closed loop (Batra et al. 2020). Recent zero-shot navigators delegate the perceptual part of this loop to large vision-language models (VLMs): a panoramic observation is described in language, a heading is selected, and a value map that scores candidate directions is updated at every step (Long et al. 2024; Yokoyama et al. 2024; Zhou et al. 2023). These methods perform well on standard benchmarks, where the input is clean. In real deployments, however, the camera rarely delivers clean frames: corridors are dimly lit, lenses fog up, and motion blurs the image. Under these conditions, the same agents lose much of their success rate: in our measurements, low light alone halves the success rate of a representative zero-shot navigator.

A natural way to address this problem is to treat it as an image-quality issue and insert a restoration front-end, such as low-light enhancement or denoising, between the camera and the policy (Guo et al. 2020; Wu et al. 2022). The underlying assumption is that once the corrupted frame is made to look clean again, the downstream navigator will behave as if the degradation were absent. Whether this assumption holds, however, has not been examined directly for VLM-driven navigators. We find that it does not, and we use a controlled diagnosis to show *why*. This is a diagnosis-and-benchmark paper: our contribution is the measurement instrument and the resulting map of what does not work, not a new agent.

A degradation-robust ObjectNav benchmark. We first build an evaluation substrate for this question: a degradation-robust ObjectNav benchmark on a representative zero-shot backbone (InstructNav (Long et al. 2024)) in Habitat-Matterport 3D (Ramakrishnan et al. 2021; Yadav et al. 2023). It applies structure-preserving corruptions (low light, haze, motion blur, sensor noise) across severities, fixes the same episodes across methods through a same-seed same-episode paired protocol that supports McNemar tests, and equips the agent with a training-free blind per-step reliability estimator. Every quantity below is measured on the unmodified agent at the $N=300$ main-split scale, so the diagnosis does not depend on any proposed fix. Sample size matters at this scale: between $N=40$ and $N=150$ the measured per-condition restoration effect *changes sign*.

Degradation breaks commitments, not perception. Three observations emerge (Figure 2). First, degradation does reduce success: under low light the success rate falls by roughly half. Second, the bottleneck is not appearance: a restoration step that raises image fidelity by 8.4 dB PSNR leaves the success rate within seed noise, and a per-frame oracle that picks the best of the degraded and restored views does

no better than blanket restoration. Third, degradation does not change *whether* the agent acts. We call the moment the agent irreversibly declares the target found—the detector’s found flag flipping on a single frame—a *commitment*; the commit rate ($\sim 93\%$) and the pure-exploration-failure rate ($\sim 6\%$) are essentially unchanged from clean, but degradation corrupts the *quality* of these irreversible commitments. Decomposing the outcome as $SR = \text{reachability} - \text{commit/stop loss}$, the agent walks within one metre of the goal in 62% of episodes yet succeeds in only 36%; it commits at a frame reliability of 0.39 versus 0.95 on clean input, only 40% of its commitments lead to success, and once committed it stays locked for ~ 12 steps. The behaviour resembles a person who, catching a brief glimpse of a doorway, strides toward it and keeps going long after it proves wrong: the agent does not fail because it cannot see; it fails because it makes an irreversible decision on a single unreliable frame and cannot revoke it.

Negative results that localize the bottleneck. The diagnosis predicts that perception-layer fixes should not help, and we confirm this at $N=300$: blanket restoration moves SR by only $+0.04$ (McNemar $p=0.18$), and even a non-deployable per-frame appearance oracle—an upper bound on any test-time arbitration over appearance—caps at $+0.057$. A broader sweep across six causal families (appearance, where-to-go, commit timing, forced commitment, temporal and multi-view aggregation, and a depth-based cross-modal verifier) finds that every *deployable* family—one that consults no privileged information at test time—either lands at a non-significant ~ 0 or *significantly hurts* success (Tab. 1), indicating an irreducible commitment error that no single test-time channel can recover.

Contributions.

- **Benchmark.** A degradation-robust ObjectNav benchmark on a zero-shot VLM backbone: structure-preserving corruptions across severities, a same-seed paired protocol with McNemar exact tests and bootstrap CIs, and a training-free blind per-step reliability estimator.
- **Diagnosis.** A method-independent mechanism analysis at $N=300$ that reframes degradation-induced failure as irreversible, low-reliability commitments, quantified by a reachability/commit-loss decomposition.
- **Negative results.** A sweep of six causal intervention families in which every deployable family caps at zero or backfires ($p<0.05$), while the non-deployable per-frame appearance oracle caps at $+0.057$; we further show that per-condition restoration effects can sign-flip between $N=40$ and $N=150$, so under-powered evaluations can conclude the wrong direction.
- **Replication.** A cross-corruption replication on the same InstructNav+Qwen2-VL-7B pipeline (low-light, motion-blur, fog, Gaussian noise) that reproduces the same non-significant fingerprint, with $|\Delta SR| \leq 0.05$ and sign-flipping across cells, establishing that the effect is not tied to any specific corruption channel. We release all settings, seeds, and negative results.

Related Work

Zero-shot object navigation with foundation models. Object-goal navigation (Batra et al. 2020) and vision-and-language navigation (Anderson et al. 2018b; Krantz et al. 2020; Khanna et al. 2024) are typically learned with reinforcement or imitation (Wijmans et al. 2020; Ramrakhya et al. 2023), which generalizes poorly to unseen distributions. Recent zero-shot systems instead build on foundation models. One family guides frontier exploration with language priors (Chaplot et al. 2020; Zhou et al. 2023; Yu, Kasaei, and Cao 2023); another fuses open-vocabulary detectors into a value map for goal reasoning (Yokoyama et al. 2024; Gadre et al. 2023); a third lets a VLM read a panoramic observation and directly emit a heading (Zhou, Hong, and Wu 2024; Long et al. 2024). We build on InstructNav (Long et al. 2024), a representative agent that couples a local VLM, an open-vocabulary detector (Wu et al. 2024), and a value-map frontier explorer. These works assume clean observations; we instead study, on the same backbone, *where* such an agent breaks when its input is degraded.

Robustness of embodied agents to visual corruption. RobustNav (Chattopadhyay et al. 2021) first benchmarked navigation under visual and dynamics corruptions and reported large drops, motivating task-oriented data augmentation (Maksymets et al. 2021) and domain adaptation. A common alternative is to restore the input before it reaches the policy via low-light enhancement (Guo et al. 2020; Wu et al. 2022) or denoising. These approaches operate at the pixel layer and aim to recover clean appearance. We show that, for a VLM-driven navigator, restoring appearance does not restore navigation: an 8.4 dB PSNR gain leaves SR within seed noise, and a per-frame oracle that always picks the better-looking view does no better. Our benchmark is designed to expose this gap, with a paired protocol and severity sweeps in the style of corruption robustness benchmarks (Hendrycks and Dietterich 2019).

Uncertainty, commitment, and executive control. Knowing when a model should not be trusted is central to safe autonomy (Abdar et al. 2021); blind image-quality assessment estimates degradation without a reference (Su et al. 2020) and active perception decides where to look in order to reduce uncertainty (Ramakrishnan, Jayaraman, and Grauman 2021). A complementary direction studies the *executive* layer of zero-shot ObjectNav: *when* to commit, stop, and recover from a bad commitment. Our diagnosis is consistent with this literature but localizes the problem more sharply: we use a training-free per-step reliability estimate purely as a *measurement instrument*, and we evaluate deployable commitment-timing and temporal/multi-view aggregation interventions head-on, each of which leaves success unchanged or lowers it on a frozen, clean-trained backbone (Tab. 1).

Evaluation of stochastic agents. VLM-driven agents are non-deterministic, and single-run point estimates of embodied performance are known to be unreliable (Anderson et al. 2018a). We therefore report multi-seed pooled success, paired per-episode differences, McNemar’s exact test, and bootstrap confidence intervals at $N=300$, and treat small-sample point

estimates only as pilot runs; the Experiments section quantifies how badly small samples mislead in this setting. Figure 1(e) visualizes the resulting coverage map of our paper against the families above.

Why Navigation Fails Under Degradation

A single episode previews the phenomenon this section quantifies. Asked to find a toilet under low light, the baseline agent navigates to within 0.02 m of it, never declares it found, wanders off, and the episode ends in failure (Figure 5c). The agent saw enough to get there; what failed was the decision to stop. This section shows that the pattern is systematic.

We analyze how visual degradation causes a zero-shot navigator to fail. Every quantity is measured on the unmodified backbone, so that the analysis does not depend on any component of our method. We organize the study as five questions (M0–M4): whether degradation hurts at all (M0), whether the cause is appearance (M1), what failures look like (M2), whether reliability is informative about commitments (M3), and what triggers a commitment (M4).

Setup. We run InstructNav on HM3D ObjectNav `val_mini` pooled over seeds $\{0, 1, 2\}$ ($n=90$, bootstrap 95% CIs, $B=5000$). The primary degradation is low-light at severity 4, drawn from a 13-type ImageNet-C-style suite (Hendrycks and Dietterich 2019). The $n=90$ study below establishes the qualitative chain; the Experiments section re-measures decisive quantities at $N=300$.

M0–M1: Degradation hurts, but appearance is not the lever. Low-light lowers SR from 0.433 (CI [0.333, 0.533]) to 0.222 ([0.144, 0.311]) at $n=90$; the paired drop is -0.211 with CI $[-0.322, -0.100]$, excluding zero. Motion-blur gives -0.056 with CI $[-0.167, +0.056]$ and does not exclude zero, so we use low-light as the primary condition. Figure 2a shows *where* the drop lives at the paired $N=300$ scale: the per-episode minimum distance to the goal is nearly unchanged (median $0.11 \rightarrow 0.15$ m)—the agent still gets to the goal—yet 48 episodes flip to failure. Adding a test-time restoration arm raises PSNR on low-light from 7.4 to 15.9 dB (+8.4 dB) yet SR moves only $0.233 \rightarrow 0.300$, inside the ± 0.10 seed-noise band: making the image look clean does not make the agent navigate.

M2: Failures are irreversible commitments. To characterize the failures, we distinguish *walk-away* failures (the agent ends far from a reachable goal and never returns) from *false stops* (premature, false-positive stops). A walk-away is *irreversible* when the trajectory was once closer to the goal but then moved away and never came back. Degradation shifts the composition of failures toward walk-aways: among failed episodes, the walk-away fraction rises from 78% on clean input to 88% under low-light, and the irreversible fraction from 73% to 79% (Figure 2b). The false-stop rate—the fraction of executed stops that are false positives—rises in parallel, from 0.51 to 0.78. Moreover, the reliability at the turning point t^* , where d_t switches from decreasing to increasing, drops from 0.93 on clean input to 0.40. The failures are thus not a matter of insufficient exploration; they are early, confident commitments made when perception is untrustworthy.

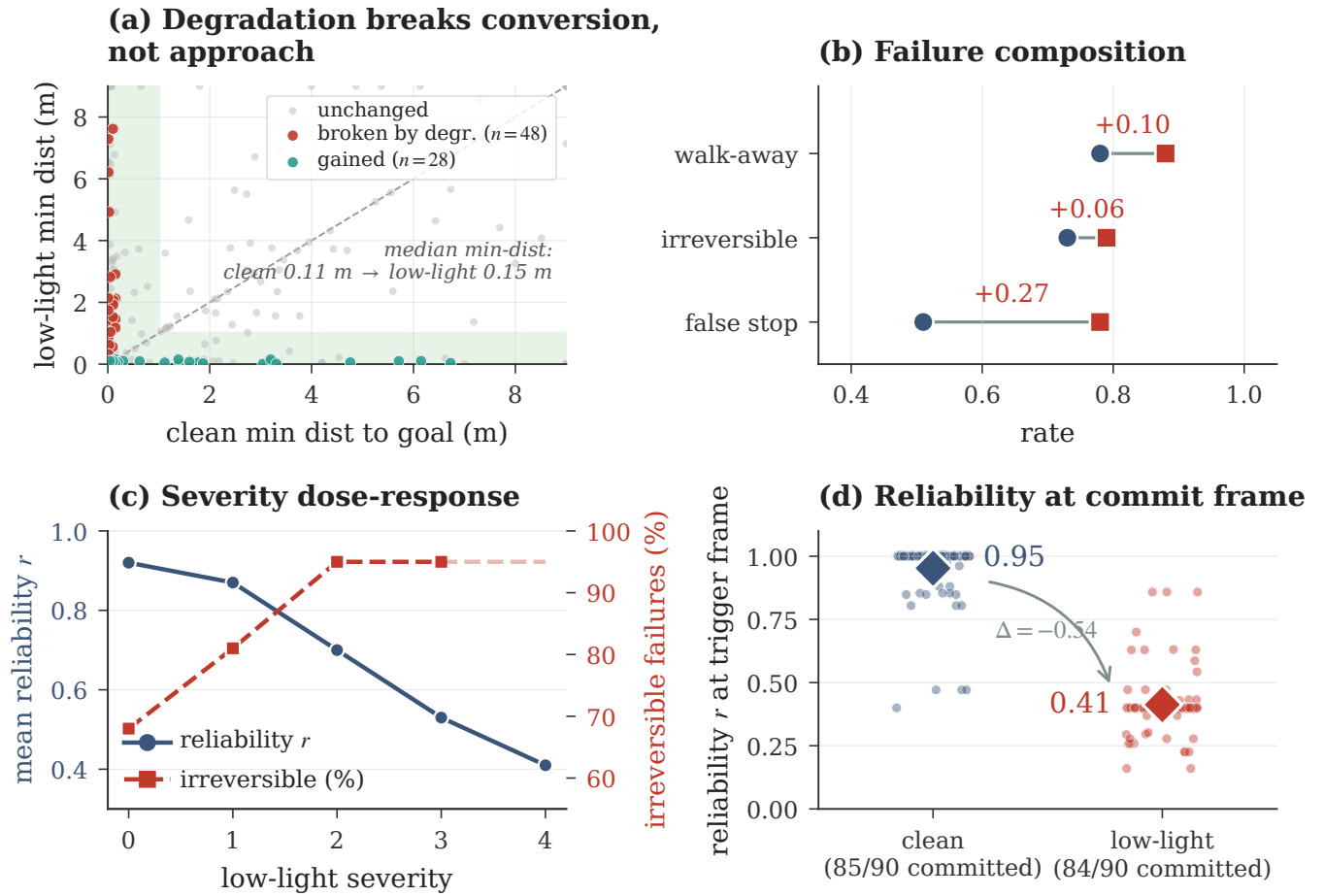


Figure 2: **Method-independent motivation study (baseline InstructNav).** (a) Per-episode paired minimum distance to goal on the *same* 300 episodes run clean and under low-light s4: the approach geometry barely moves (median min-dist 0.11 $\rightarrow$ 0.15 m) yet 48 episodes flip to failure (red)—degradation breaks the reach-to-success *conversion*, not the approach. (b) M2 ($n=90$): degradation shifts failures toward walk-aways and irreversible failures, and false stops nearly double (walk-away and irreversible: fraction of failed episodes; false stop: fraction of executed stops that are false positives). (c) M3: as severity rises, mean reliability falls and the irreversible fraction saturates at 95%. (d) M4: the deciding commitment fires on a single low- r frame (mean r drops 0.95 $\rightarrow$ 0.41); once committed the agent stays locked for ~ 21 steps. Coloured dots in (d) jitter the underlying per-commit r distribution; diamonds are the means. N counts episodes in which a commit fired (85 and 84 of the 90 per condition; the remainder never committed).

M3–M4: Reliability is informative; a single low- r frame triggers the commitment. The reliability estimate carries a usable signal. On clean input, bucketing commitments by r shows net geodesic progress over the next five steps rising monotonically with r (0.133/0.154/0.245 for low/mid/high), and across severities 0–4 the mean r falls monotonically (0.92 $\rightarrow$ 0.87 $\rightarrow$ 0.70 $\rightarrow$ 0.53 $\rightarrow$ 0.41) while the irreversible fraction saturates at 95% by severity 3 (Figure 2c). The commitment itself is a single-frame event: the detector’s found flag flips on one frame, not after temporal consensus. Across conditions, the r of that triggering frame falls from 0.95 on clean to 0.41 under low-light (85 and 84 of the 90 episodes per condition committed at least once), and once made the commitment persists for ~ 21 steps (Figure 2d). The entry point

of the failure is an irreversible decision on one unreliable frame.

M5: A reachability/commit-loss decomposition at the $N=300$ scale. The M -study above uses $n=90$; we now confirm the mechanism at the $N=300$ main-split scale, where the baseline success rate is 0.36 rather than the 0.22 of *valmini*. We decompose the outcome as $\text{SR} = \text{reachability} - \text{commit/stop loss}$, where reachability is the fraction of episodes whose trajectory passes within 1.0 m of the goal (an upper bound achievable by a perfect stop policy on the *actual* trajectory, requiring no re-run; the threshold is calibrated from the p_{90} stopping distance of clean successes, 0.15 m). On low-light at severity 4, the baseline reaches within 1.0 m in 62.3% of episodes but succeeds in only 35.8%, a commit/stop loss of 26.5 points: in one episode

out of four the agent passes the goal and fails to convert. Crucially, degradation does not suppress *action*: the commit rate is 93% under low-light versus 94% on clean input, and the pure-exploration-failure rate is $\sim 6\%$ at both ends. What collapses is commitment *quality*: the reliability at the committing frame falls from 0.95 to 0.39, the false-commit rate rises from 55% to 76%, only 40% of commitments lead to success, and the agent stays locked for ~ 12 steps (~ 21 in the $n=90$ study; the lock is shorter on the easier main split but present at both scales). The same decomposition holds for Gaussian noise (reachability ~ 0.59 , the loss again dominant), so the pattern is not specific to low light.

Summary. The observations form a consistent chain: degradation does not blind the agent and does not stop it from acting; it leads the agent to stake irreversible heading and stop commitments on single, low-reliability frames, so the resulting loss sits in the reach-to-success conversion rather than in perception or exploration. The diagnosis predicts that interventions on appearance or on where-to-go should not move success, because they never touch the commitment conversion. The Method section turns this into candidate interventions; the Experiments section confirms the prediction across a six-family sweep in which no deployable arm moves success and several lower it.

From Diagnosis to Candidate Interventions

The diagnosis specifies both *what* fails (the quality of irreversible commitments) and *when* (when the committing frame is unreliable), and makes a falsifiable prediction: interventions on image appearance or on where-to-go should not move success, because the loss lives in the reach-to-success conversion. To test this prediction we construct the most natural candidate interventions the diagnosis suggests, spanning the pixel, perception, and decision layers. A per-frame appearance oracle upper-bounds any test-time arbitration over appearance. The instrument behind the test (Figure 4) is a paired protocol that exposes the same frozen backbone to clean and degraded versions of identical episodes and measures the per-episode difference.

Per-Step Reliability Estimation

We map each RGB frame to a scalar reliability $r \in [0, 1]$ using a blind detector bank that needs no reference frame or training. We combine cues that respond to structure-preserving corruptions: mean luma for darkness, global contrast and std. deviation for a haze veil, Laplacian energy for blur, a dark-channel statistic for fog, and a noise estimate for sensor noise. Each cue is calibrated once on a probe set so clean frames score $r \approx 1$ and corruptions lower r monotonically (Fig. 3): clean 1.00, low-light 0.65, motion-blur 0.47, fog 0.36, Gaussian noise 0.20. Throughout, r serves as an *instrument*: it lets candidates gate on perceived trustworthiness, but is not itself a contribution.

Candidate 1 (Appearance): Blanket and Oracle-Gated Restoration

The default remedy treats degradation as an image-quality problem and restores the frame before the policy sees it.

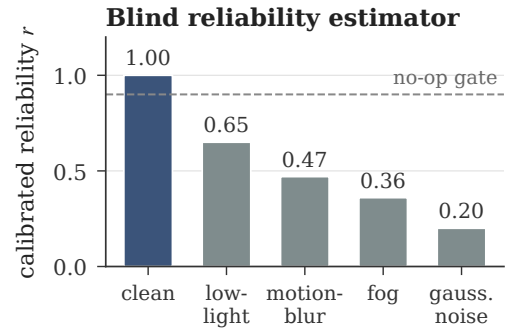


Figure 3: **Blind reliability estimator** r on the calibration probe set. Clean frames score $r \approx 1$; each structure-preserving corruption lowers r monotonically. Dashed line is the $r=0.9$ no-op gate above which all reliability-gated candidates reduce to the baseline.

We include two variants. *Blanket restoration* (RES) always applies a low-light/denoising front-end. To upper-bound *any* test-time arbitration over appearance, we also define a *per-frame appearance oracle* (ORACLEGATE) that, at every step, is allowed to pick whichever of the degraded and restored views yields the better navigation outcome. ORACLEGATE is not deployable—it consults privileged information—but it bounds the best a perfect router over restored versus degraded appearance could ever achieve. If even ORACLEGATE does not beat blanket restoration, no appearance-level decision can.

Candidate 2 (Where-to-Go): Reliability-Gated Reweighting

If appearance is not the lever, the diagnosis points to the decision layer. A value-map navigator scores candidate headings by combining a *language-semantic term* $v_{\text{vlm+sem}}$ (which a corrupted frame misleads) with a geometric frontier-coverage term v_{frontier} . The reliability-gated reweighting candidate (M1g) takes a convex mixture $v(a) = (1-\beta)v_{\text{vlm+sem}}(a) + \beta v_{\text{frontier}}(a)$, with the mixing weight $\beta(r)$ gated by reliability: $\beta=0$ above $\tau_{\text{gate}}=0.9$, and $\beta = \min(\beta_{\text{max}}, \alpha(1-r))$ below it ($\alpha=1$, $\beta_{\text{max}}=0.6$), so on clean input M1g reduces exactly to the backbone.

Together these candidates probe pixels (RES, OracleGate) and where-to-go (M1g); ORACLEGATE is an upper bound rather than a method, so the collective failure to move success would be a statement about *where the headroom is*, not about weak baselines. The Experiments section confirms this and sweeps four further families (commitment timing/revocation, forced commitment, temporal/multi-view aggregation, and a depth-based multimodal probe), so the headroom claim is bounded across the full intervention space.

Experiments

We now test the prediction of the diagnosis at scale. We first describe the paired protocol, then report (i) the degradation effect that defines the benchmark, (ii) two negative results that localize the bottleneck, (iii) a mechanism analysis

that explains *why* the candidates cap, (iv) a sweep showing the bottleneck resists the full intervention space, and (v) a methodological warning about sample size.

Protocol

We evaluate on HM3D ObjectNav with the InstructNav backbone (Long et al. 2024), using an open-source Qwen2-VL-7B-Instruct (Wang et al. 2024) as the reasoning VLM (deterministic decoding on a single RTX 5090; no external API), the GLEE open-vocabulary detector (Wu et al. 2024) for segmentation, and InstructNav’s value-map frontier explorer for goal selection. Choosing an open-weight backbone lets every frame of every episode be re-run bit-identical from the released seeds and prompts, which is the reproducibility bar we hold ourselves to for a negative-result release. Conclusions are drawn at the $N=300$ main-split scale; smaller runs are used only as pilot runs and are never mixed with $N=300$ numbers. Each arm is run against the identical backbone on the same episodes, so comparisons are paired. We report success rate (SR) and SPL, the paired per-episode difference ΔSR , McNemar’s exact test on the discordant pairs, and a bootstrap 95% CI over the paired differences ($B=5000$). Figure 4 (top) summarizes the instrument. The primary degradation is low-light at severity 4, which darkens the image while preserving geometry (Fig. 4, bottom). We avoid single-run point estimates because the clean SR of the backbone alone ranges over 0.27–0.47 across seeds.

The Degradation Effect (Benchmark)

Structure-preserving degradation produces a large, real drop in navigation success, which is the effect the benchmark is built to study. Pooling three seeds ($n=90$), low-light lowers SR from 0.433 (95% CI [0.333, 0.533]) to 0.222 ([0.144, 0.311]); the paired drop is -0.211 with CI $[-0.322, -0.100]$, excluding zero. At the $N=300$ main split the low-light baseline SR is 0.37. Motion-blur at the same severity gives -0.056 with CI $[-0.167, +0.056]$, which does not exclude zero, so we use low-light as the primary condition and report motion-blur as a non-significant case. The benchmark applies the same protocol across low-light, fog, motion-blur, and Gaussian noise at multiple severities.

Two Negative Results at $N=300$

The diagnosis predicts that appearance-level and where-to-go interventions should not recover success, and Table 1 confirms this at the $N=300$ main split. Blanket restoration (RES) raises low-light SR from 0.370 to 0.410, a paired ΔSR of $+0.040$ that is not significant ($p=0.18$); the per-frame appearance oracle (ORACLEGATE), an upper bound on any test-time arbitration over appearance, reaches only 0.427 ($+0.057$, $p=0.04$), statistically indistinguishable from blanket restoration ($+0.017$ over RES). Reliability-gated where-to-go reweighting (M1g) also reaches $+0.040$ ($p=0.20$). Every candidate converges to the same non-significant $\sim+0.04$, and the oracle in particular leaves no room: no router over restored versus degraded views can exceed it.

Figure 5 shows what these aggregates look like at the episode level, one paired episode per outcome bucket: an oracle *rescue*, an oracle *break* (greedy per-frame view selection

walks the agent away from a goal RES had reached), and a *tied* failure in which the baseline parks 0.02 m from the goal yet never commits—reach without success, the signature of the commitment bottleneck. Figure 6 shows the same failure modes spatially: top-down maps of paired episodes in which all three arms start from the identical pose and the trajectories diverge only through the intervention.

Mechanism: Why the Candidates Cap

The decomposition $\text{SR} = \text{reach} - \text{commit/stop loss}$ explains the ceiling: across B0, RES, and M1g, reach (~ 0.62) and commit/stop loss (~ 0.25) are nearly invariant, so no candidate changes whether the agent reaches the goal or converts a reach into a stop. The per-episode logs localize *where* this invariance lives (Fig. 7). First, the distribution of the distance at which the agent *first commits* is unchanged by any intervention—the B0, RES, and even OracleGate ECDFs coincide, with $\sim 70\%$ of first commits firing more than 3 m from the goal (Fig. 7a). Second, RES shifts the reliability *distribution* at those commit frames from a median of 0.40 to 1.00, so the frames now look trustworthy, yet the probability that a commit converts to success moves only $0.42 \rightarrow 0.43$ (Fig. 7b). Third, the episodes whose outcome does flip under RES were already at the 1 m success margin under B0 (median rescued episode: 1.1 m), so the intervention perturbs threshold episodes rather than repairing far failures (Fig. 7c). Restoring perception without repairing decision is precisely the 26-point reach-to-success gap that no candidate touches, which is exactly the $\sim+0.04$ ceiling observed in Table 1.

The Bottleneck Resists the Full Intervention Space

If the loss were a single mislocated module, *some* layer-specific fix would move it. We therefore sweep six causal families (Table 1, visualized in Fig. 1d): appearance (restoration and its per-frame oracle upper bound), where-to-go reweighting, commit-timing revocation, forced commitment, temporal/multi-view aggregation, and a depth-based multi-modal probe (Sec. *Probing the Multimodal Direction*). All non-oracle arms below the main split are scored at a common $N=150$ on the same low-light sev 4 seed 0 val episodes and paired against the same $N=150$ baseline, so all comparisons are exactly on the same episodes. Every *deployable* family either lands at the same non-significant ~ 0 or *significantly hurts* success: REVOKE, MUAP and DXCV lower paired SR by 0.12–0.17 at $p<0.05$ (DXCV at $p<0.001$); the only arm clearing $p<0.05$ with a *positive* ΔSR is the non-deployable appearance oracle, and even it caps at $+0.057$. Crucially, accumulating evidence across frames, views, or modalities does not just fail to recover the decision—it can *amplify* the wrong one, because the degradation removes the information that distinguishes the goal at the decision point rather than adding independent noise that averages out.

Probing the Multimodal Direction

A natural worry is that the bottleneck is specifically an RGB problem and that adding a degradation-invariant sensing channel would dissolve it. We test this with a train-free probe, DXCV (Depth Cross-Modal Verifier): when the

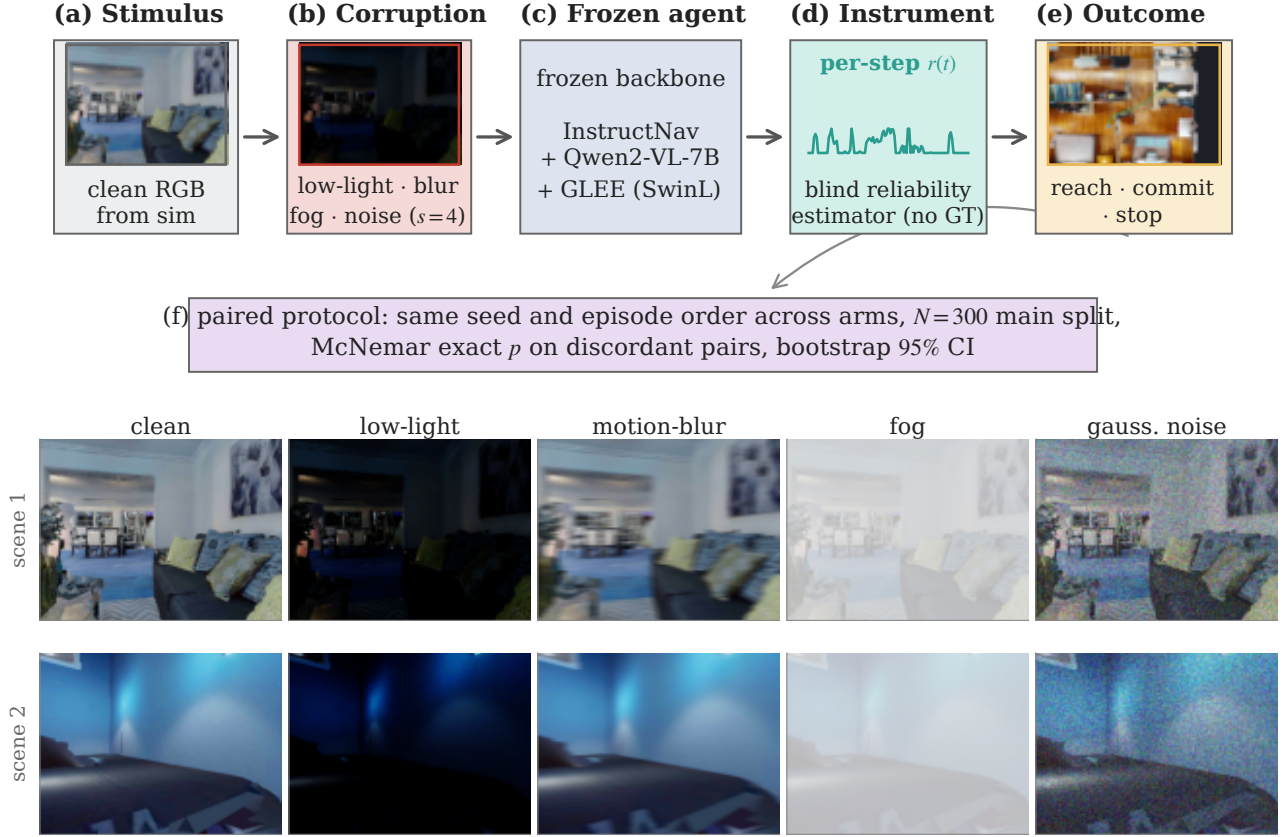


Figure 4: **The diagnosis instrument and its stimuli.** *Top*: a frozen InstructNav backbone is exposed to identically-seeded clean and degraded ObjectNav episodes; a blind per-step reliability estimator instruments the policy without changing it; outcomes are reach (within 1 m), commit, and stop; paired McNemar exact p on the discordant pairs is the unit of comparison. *Bottom*: the four corruption channels at the severity used in all main results—two HM3D val scenes rendered clean and under each corruption at severity 4: low-light (gamma), motion-blur (kernel 25), fog (intensity 0.8), Gaussian noise ($\sigma=0.35$). PSNR vs. clean drops from >30 dB to 7–16 dB depending on channel. All corruptions preserve scene geometry; what varies is the photometric evidence available to the RGB channel.

VLM proposes `Flag=True` at a low-reliability frame, a low-threshold GLEE pass locates the target mask and the depth sensor must agree—masked depth median in $[0.3, 3.5]$ m and IQR at most 0.6 m; otherwise the commit is vetoed and search resumes. Habitat’s depth channel is rendered from geometry rather than light, so it is invariant to low-light corruption. Paired against the same $N=150$ baseline (Table 1): $\Delta\text{SR}=-0.173$ (McNemar $p<0.001$), a *strongly negative effect*. Vetoing geometrically implausible commits on a frozen agent removes a small number of true positives along with the false ones; without having *learned* the joint RGB+depth representation the verifier cannot tell them apart, and the net effect of bolting an extra modality onto a clean-trained policy is to *lower* SR rather than raise it. A multimodal *remedy*, as opposed to a multimodal *probe*, therefore requires a detector that has learned to fuse RGB with depth under degraded RGB—the training-side fix the Conclusion advocates.

Sample Size and Scope

The per-condition restoration effect is small and sample-sensitive: on Gaussian noise, an $N=40$ pilot yields $\Delta\text{SR}=-0.125$ (RES apparently *hurts*), while a matched $N=150$ run collapses to $\Delta\text{SR}=+0.000$ —the pilot’s negative direction does not survive scale. We therefore report all conclusions at $N=300$, and treat smaller runs as pilot runs only (Appendix A). The main-split numbers use the InstructNav planner with Qwen2-VL-7B, with low-light as the primary degradation. The next subsection extends the headline cell to the other three corruption channels at matched $N=150$ paired scale.

Cross-Corruption Replication

A reviewer-anticipated concern is that the bottleneck is a low-light artefact: motion-blur, fog, or noise might preserve enough photometric evidence that restoration *does* recover success. We test this directly by holding the

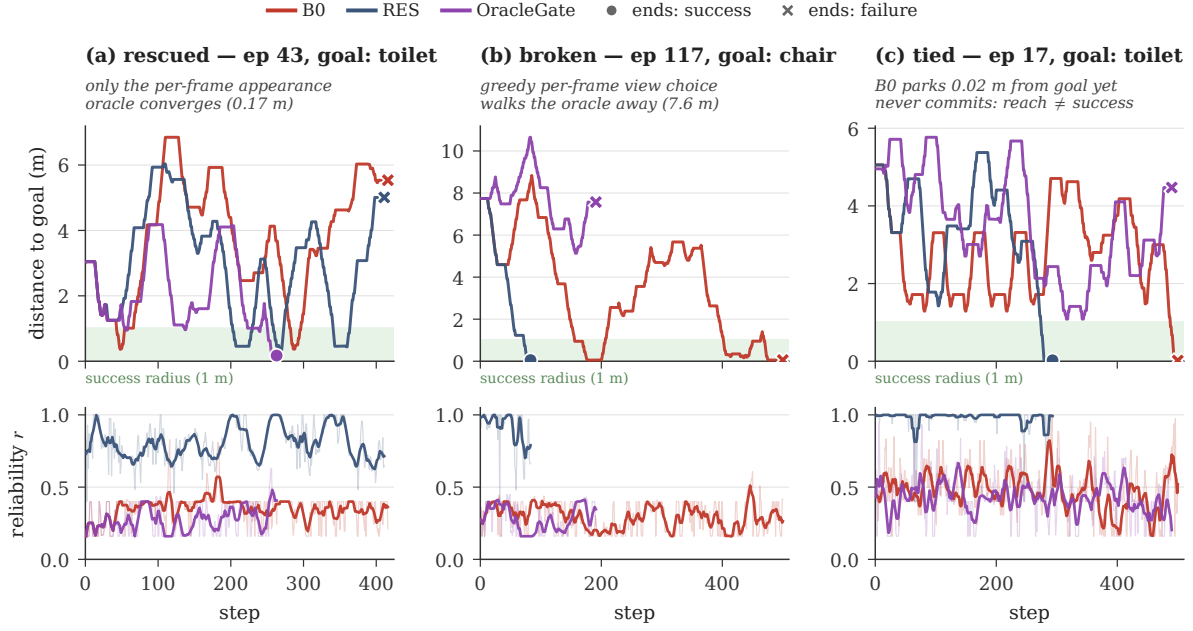


Figure 5: **Episode-level decision traces from the paired $N=300$ low-light cell.** Top: distance to goal per step for B0, RES, and ORACLEGATE on the *same* episode; markers show terminal success (●) or failure (×); the green band is the 1 m success radius. Bottom: per-step blind reliability r (faint: raw; solid: 9-step moving average). **(a) rescued:** only the non-deployable per-frame appearance oracle converges. **(b) broken:** the same oracle, by greedy per-frame view choice, abandons a goal RES had secured. **(c) tied:** B0 touches 0.02 m from the goal yet never commits—reach and success decouple. Note in all three cases RES (blue) sustains high r : perception is confidently restored while the decision outcome is unchanged.

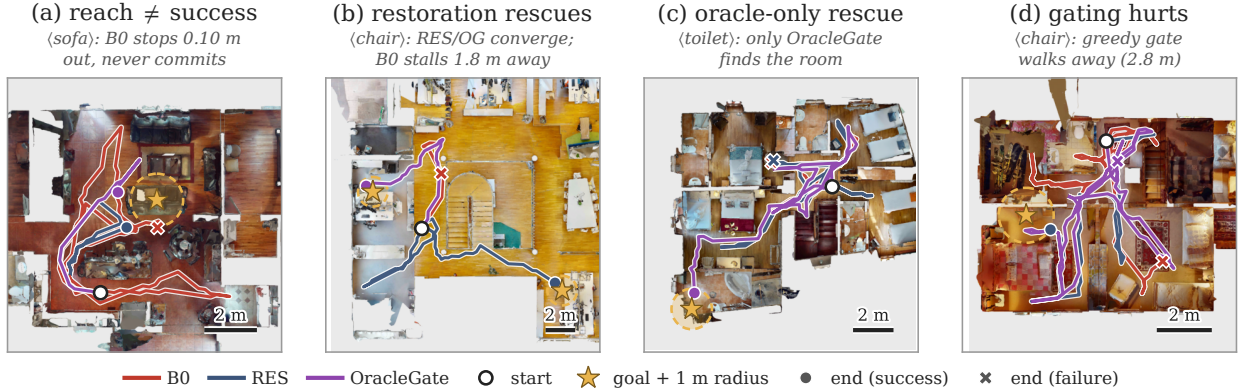


Figure 6: **Top-down trajectories of paired episodes (low-light sev. 4).** Backgrounds are bird’s-eye orthographic renders of the actual HM3D scenes (ceiling clipped); trajectories are the logged per-step world positions, so curves sit exactly on the floors they were walked on. All arms share the identical start pose and episode; stars mark the goal instances with the 1 m success radius. **(a)** B0 (red) passes 0.10 m from the sofa yet never issues a stop—reach without success. **(b)** restoration rescues: RES (blue) and ORACLEGATE (purple) both convert, B0 stalls 1.8 m out. **(c)** only the per-frame appearance oracle finds the target room; B0 and RES wander the same wrong wing. **(d)** the same oracle, choosing views greedily per frame, abandons a chair that RES secures and ends 2.8 m away. The interventions reroute *where* the agent goes far less than they change *whether* it commits.

InstructNav+Qwen2-VL-7B pipeline bit-identical across the four channels—mapper, depth sensor, GLEE detector, frontier planner, prompt, decoding seed, episode list, split, and severity all frozen—and only swapping the corruption. Table 2 reports the paired benchmark. No cell reaches significance at $\alpha=0.05$; the sign of ΔSR flips between motion-blur

(−0.040) and the other three channels (+0.000—+0.047), reproducing the small-sample sign-flip diagnosis of *Sample Size and Scope*. The bottleneck is thus not a property of any one corruption but of the *commitment-on-degraded-input* interface that VLM-driven planners share.

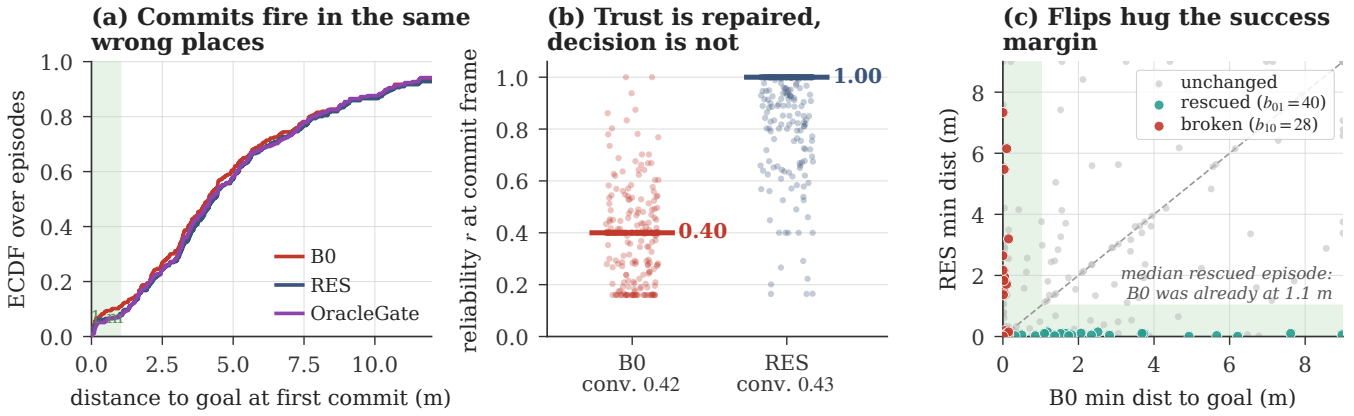


Figure 7: **Inside the commitment bottleneck (per-episode logs of the paired $N=300$ low-light runs; nothing here repeats Table 1).** (a) ECDF of the distance to goal at the *first commit event*: the B0, RES, and ORACLEGATE curves are nearly indistinguishable, and $\sim 70\%$ of first commits fire more than 3 m from the goal—no intervention changes *where* the agent decides it has found the object. (b) Blind reliability r at the commit frame, one dot per committing episode: RES shifts the median from 0.40 to 1.00, yet the probability that a commit converts to success moves only $0.42 \rightarrow 0.43$. (c) Paired per-episode minimum distance (B0 vs. RES) with outcome flips coloured: rescued episodes (teal) sit almost entirely inside or at the 1 m success band under B0 already (median 1.1 m), and broken episodes (red) mirror them—the intervention shuffles the margin rather than moving the population of far failures.

Family	Arm	What it does	SR_{B0}	SR_{arm}	ΔSR	ΔSPL	95% CI(ΔSR)	p (McN.)	N
Appearance	RES	blanket restoration front-end	0.370	0.410	+0.040	+0.032	$[-0.013, +0.093]$	0.182	300
Appearance UB	OracleGate [‡]	per-frame best-of-two view	0.370	0.427	+0.057	+0.025	$[+0.003, +0.110]$	0.043	300
Where-to-go	M1g	r -gated value-map reweight	0.370	0.410	+0.040	+0.012	$[-0.017, +0.097]$	0.201	300
Commit timing	REVOKE	revoke stalled commitments	0.380	0.260	-0.120	-0.020	$[-0.207, -0.040]$	0.006	150
Forced commit	CRV	close-range recall verifier	0.380	0.400	+0.020	+0.021	$[-0.053, +0.093]$	0.728	150
Temporal	MUAP	hold-frames + re-observe	0.380	0.260	-0.120	-0.039	$[-0.220, -0.020]$	0.022	150
Temporal	ROUTER	per-frame degradation router	0.380	0.393	+0.013	+0.031	$[-0.073, +0.100]$	0.875	150
Multi-view	FUSE	multi-view identity fusion	0.380	0.367	-0.013	+0.032	$[-0.100, +0.073]$	0.878	150
Multimodal probe	DXCV ^b	depth cross-modal veto	0.380	0.207	-0.173	-0.057	$[-0.260, -0.087]$	<0.001	150

Table 1: **Intervention sweep across six causal families.** Paired protocol on low-light sev 4, seed 0, val split. The three top rows are scored at the $N=300$ main split; the six lower rows share a common $N=150$ subset with a single shared baseline, so all comparisons are on exactly the same episodes. Bold entries are significant at $p < 0.05$ (McNemar exact). Every *deployable* family either lands at a non-significant ~ 0 or *significantly hurts* success (REVOKE, MUAP, DXCV lower paired SR by 0.12–0.17); the only arm clearing $p < 0.05$ with a *positive* ΔSR is the non-deployable per-frame appearance upper bound, which caps at +0.057. Confidence intervals are $B=5000$ bootstrap over paired episode differences. [‡]non-deployable oracle. ^btrain-free depth-geometry cross-modal veto.

A first training-side probe. If the diagnosis is right, the lever the sweep could not reach should be reachable by changing what the agent was *trained* on. As a minimal probe we train a 10^6 -parameter residual head adapter on (clean, degraded) SwinL feature pairs from scene-disjoint episodes (2k steps; Appendix H). On held-out scenes it moves degraded features 13.5% closer to their clean counterparts in L1, and its paired navigation effect ($\Delta SR = +0.040$, $N=150$, $p=0.377$) already matches the best deployable test-time arm of Table 1—while leaving the entire training recipe untouched. At pilot scale the effect is not statistically resolvable and we claim no more than direction; we report it as the baseline the released benchmark is designed to measure.

Conclusion

We asked why VLM-driven navigation agents fail under visual degradation and showed that the common remedy—restoring the input—is aimed at the wrong layer: the agent keeps acting and keeps seeing, but commits irreversibly on unreliable frames, leaving a 26-point reach-to-success gap that no deployable test-time intervention closes. The sweep, its oracle upper bound, and the cross-corruption replication together locate this loss at the commitment interface itself rather than in any specific corruption channel or module. The remaining headroom is accessible only by changing what the agent was *trained* on: degradation-aware training and adaptation that keep perception and commitment discriminative under corruption, and, where multi-modal sensing

Corruption	SR _{B0}	SR _{RES}	Δ SR	95% CI(Δ SR)	p (McN.)
low-light	0.387	0.433	+0.047	$[-0.027, +0.127]$	0.311
motion-blur	0.367	0.327	-0.040	$[-0.133, +0.047]$	0.461
fog	0.407	0.433	+0.027	$[-0.053, +0.107]$	0.627
gauss. noise	0.253	0.253	+0.000	$[-0.073, +0.073]$	1.000

Table 2: **Cross-corruption paired benchmark** ($N=150$, seed 0). InstructNav + Qwen2-VL-7B on HM3D val, held bit-identical across the four corruption channels (severity 4); the only comparison is blanket RGB restoration (RES) against the frozen baseline (B0), the intervention a reviewer would first ask about (Δ SPL is similarly null on every cell, $|\Delta$ SPL| ≤ 0.03). All four cells stay non-significant at $\alpha=0.05$, and the sign of Δ SR flips between motion-blur (−) and the other three (+)—matching the small-sample sign-flip diagnosis in *Sample Size and Scope*. The commitment-on-degraded-input fingerprint is thus not tied to any single corruption channel.

is available, detectors that have *learned* to fuse RGB with depth under degraded input. A first pilot in this direction—a 10^6 -parameter feature adapter trained on scene-disjoint (clean, degraded) pairs—already matches the best deployable test-time arm while measurably re-aligning degraded features (Appendix H); the effect is not yet statistically resolvable, but it is the qualitative direction the diagnosis predicts. We release the benchmark, paired protocol, and all negative results as an instrument for measuring such retrained agents under the same paired, adequately powered conditions.

Limitations. Our diagnosis is bounded by its substrate: one simulator (Habitat HM3D), one task (ObjectNav), one open-weight VLM backbone (Qwen2-VL-7B), and synthetic ImageNet-C-style corruptions; real sensor degradation may add artefacts (rolling shutter, auto-exposure hunting) we do not model. The intervention sweep covers six causal families but cannot exhaust the design space of test-time fixes, so our negative results bound the families we probed, not every conceivable patch. Finally, the reliability estimator is an analysis instrument calibrated on our probe set; we do not claim it transfers unchanged to other embodiments.

References

Abdar, M.; Pourpanah, F.; Hussain, S.; Rezazadegan, D.; Liu, L.; Ghavamzadeh, M.; Fieguth, P.; Cao, X.; Khosravi, A.; Acharya, U. R.; Makarenkov, V.; and Nahavandi, S. 2021. A review of uncertainty quantification in deep learning: Techniques, applications and challenges. *Information Fusion*, 76: 243–297.

Anderson, P.; Chang, A.; Chaplot, D. S.; Dosovitskiy, A.; Gupta, S.; Koltun, V.; Kosecka, J.; Malik, J.; Mottaghi, R.; Savva, M.; and Zamir, A. R. 2018a. On Evaluation of Embodied Navigation Agents. *arXiv preprint arXiv:1807.06757*.

Anderson, P.; Wu, Q.; Teney, D.; et al. 2018b. Vision-and-Language Navigation: Interpreting Visually-Grounded Navigation Instructions in Real Environments. In *CVPR*.

Batra, D.; Gokaslan, A.; Kembhavi, A.; Maksymets, O.; Mottaghi, R.; Savva, M.; Toshev, A.; and Wijmans, E. 2020. ObjectNav Revisited: On Evaluation of Embodied Agents Navigating to Objects. *arXiv preprint arXiv:2006.13171*.

Chaplot, D. S.; Gandhi, D.; Gupta, A.; and Salakhutdinov, R. 2020. Object Goal Navigation using Goal-Oriented Semantic Exploration. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Chattopadhyay, P.; Hoffman, J.; Mottaghi, R.; and Kembhavi, A. 2021. RobustNav: Towards Benchmarking Robustness in Embodied Navigation. In *IEEE/CVF International Conference on Computer Vision (ICCV)*.

Gadre, S. Y.; Wortsman, M.; Ilharco, G.; Schmidt, L.; and Song, S. 2023. CoWs on Pasture: Baselines and Benchmarks for Language-Driven Zero-Shot Object Navigation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

Guo, C.; Li, C.; Guo, J.; Loy, C. C.; Hou, J.; Kwong, S.; and Cong, R. 2020. Zero-Reference Deep Curve Estimation for Low-Light Image Enhancement. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

Hendrycks, D.; and Dietterich, T. 2019. Benchmarking Neural Network Robustness to Common Corruptions and Perturbations. *International Conference on Learning Representations (ICLR)*.

Khanna, M.; Ramrakhya, R.; Chang, M.; et al. 2024. GOAT-Bench: A Benchmark for Multi-Modal Lifelong Navigation. In *CVPR*.

Krantz, J.; Wijmans, E.; Majumdar, A.; Batra, D.; and Lee, S. 2020. Beyond the Nav-Graph: Vision-and-Language Navigation in Continuous Environments. In *ECCV*.

Long, Y.; Cai, W.; Wang, H.; Zhan, G.; and Dong, H. 2024. InstructNav: Zero-shot System for Generic Instruction Navigation in Unexplored Environment. In *Conference on Robot Learning (CoRL)*.

Maksymets, O.; Cartillier, V.; Gokaslan, A.; et al. 2021. THDA: Treasure Hunt Data Augmentation for Semantic Navigation. In *ICCV*.

Ramakrishnan, S. K.; Gokaslan, A.; Wijmans, E.; Maksymets, O.; Clegg, A.; Turner, J.; Undersander, E.; Galuba, W.; Westbury, A.; Chang, A. X.; Savva, M.; Zhao, Y.; and Batra, D. 2021. Habitat-Matterport 3D Dataset (HM3D): 1000 Large-scale 3D Environments for Embodied AI. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks*.

Ramakrishnan, S. K.; Jayaraman, D.; and Grauman, K. 2021. An Exploration of Embodied Visual Exploration. *International Journal of Computer Vision (IJCV)*, 129(5): 1616–1649.

Ramrakhya, R.; Batra, D.; Wijmans, E.; and Das, A. 2023. PIRLNav: Pretraining with Imitation and RL Finetuning for ObjectNav. In *CVPR*.

Su, S.; Yan, Q.; Zhu, Y.; Zhang, C.; Ge, X.; Sun, J.; and Zhang, Y. 2020. Blindly Assess Image Quality in the Wild Guided by a Self-Adaptive Hyper Network. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

Wang, P.; Bai, S.; Tan, S.; Wang, S.; Fan, Z.; Bai, J.; Chen, K.; Liu, X.; Wang, J.; Ge, W.; Fan, Y.; Dang, K.; Du, M.; Ren, X.; Men, R.; Liu, D.; Zhou, C.; Zhou, J.; and Lin, J. 2024. Qwen2-VL: Enhancing Vision-Language Model's Perception of the World at Any Resolution. *arXiv preprint arXiv:2409.12191*.

Wijmans, E.; Kadian, A.; Morcos, A.; et al. 2020. DD-PPO: Learning Near-Perfect PointGoal Navigators from 2.5 Billion Frames. In *ICLR*.

Wu, J.; Jiang, Y.; Liu, Q.; Yuan, Z.; Bai, X.; and Bai, S. 2024. GLEE: General Object Foundation Model for Images and Videos at Scale. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

Wu, W.; Weng, J.; Zhang, P.; Wang, X.; Yang, W.; and Jiang, J. 2022. URetinex-Net: Retinex-based Deep Unfolding Network for Low-light Image Enhancement. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

Yadav, K.; Ramrakhya, R.; Ramakrishnan, S. K.; Gervet, T.; Turner, J.; Gokaslan, A.; Maestre, N.; Chang, A. X.; Batra, D.; Savva, M.; Clegg, A. W.; and Chaplot, D. S. 2023. Habitat-Matterport 3D Semantics Dataset. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

Yokoyama, N.; Ha, S.; Batra, D.; Wang, J.; and Bucher, B. 2024. VLFM: Vision-Language Frontier Maps for Zero-Shot Semantic Navigation. In *IEEE International Conference on Robotics and Automation (ICRA)*.

Yu, B.; Kasaei, H.; and Cao, M. 2023. L3MVN: Leveraging Large Language Models for Visual Target Navigation. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.

Zhou, G.; Hong, Y.; and Wu, Q. 2024. NavGPT: Explicit Reasoning in Vision-and-Language Navigation with Large Language Models. In *AAAI Conference on Artificial Intelligence*.

Zhou, K.; Zheng, K.; Pryor, C.; Shen, Y.; Jin, H.; Getoor, L.; and Wang, X. E. 2023. ESC: Exploration with Soft Commonsense Constraints for Zero-shot Object Navigation. In *International Conference on Machine Learning (ICML)*.

A Sample Size and Sign-Flip Diagnostic

The per-condition restoration effect is small and sample-sensitive: on the Gaussian-noise pilot run ($N=40$) we measure $\Delta\text{SR} = -0.125$, while the matched $N=150$ comparison of Tab. 2 collapses to $\Delta\text{SR} = +0.000$ —the pilot’s negative direction is an artefact of which five episodes the small sample draws. We therefore report all main conclusions at $N=300$ with paired tests, and warn that under-powered degradation evaluations can yield conclusions of the wrong sign. The main-split numbers use a single pipeline (InstructNav+Qwen2-VL-7B) with low-light as the primary degradation. Broadening the $N=300$ analysis across fog and motion-blur, and replicating the bottleneck on a second value-navigator (e.g. VLFM), are the natural next steps.

B Paired Protocol and Statistical Tests

The diagnosis instrument is illustrated in Figure 4 of the main paper. For each (severity, seed) pair, every arm is evaluated on the same fixed list of episodes in the same order, so that the per-episode success vector $y^{\text{arm}} \in \{0, 1\}^N$ is directly comparable across arms. The paired difference vector is $\Delta_i = y_i^{\text{arm}} - y_i^{\text{B0}}$; the reported ΔSR is the mean of Δ_i . For McNemar’s exact two-sided test we count b_{01} (B0 fails, arm succeeds) and b_{10} (B0 succeeds, arm fails) and use the exact binomial tail $p = 2 \sum_{i=0}^{\min(b_{01}, b_{10})} \binom{b_{01}+b_{10}}{i} / 2^{b_{01}+b_{10}}$, clipped to 1. Bootstrap 95% CIs on ΔSR resample $\{\Delta_i\}$ with replacement $B=5000$ times. We use a fixed numpy RNG seed of 42 across all reported bootstrap intervals.

C Hyperparameters of the Probed Interventions

Each non-baseline arm exposes the following hyperparameters; all values are fixed across episodes within an arm and were selected on the calibration probe set rather than on the val episodes used for reporting.

- **RES**: a blanket low-light restoration network, applied at every frame; no reliability gate.
- **M1g (reliability-gated reweight)**: $\alpha=1.0$, $\beta_{\text{max}}=0.6$, $\tau_{\text{gate}}=0.9$.
- **OracleGate (upper bound)**: at every step, an oracle picks whichever of the degraded or restored view yields the better navigation outcome; not deployable.
- **REVOKE**: stall-window $K=6$, displacement floor $\epsilon=0.15$, far-from-goal threshold 0.8 m, ≤ 2 revocations per episode, cooldown 2.0 s.
- **CRV (close-range recall verifier)**: rel-gate 0.90, GLEE conf 0.10, floor 0.12, mask area ≥ 5000 px, frames 1, max 3 verifications, centre tolerance 0.18.
- **MUAP**: target hold-frames 8, min frames 3, match distance 0.6 m; uap reobserve mode, $\tau_{\text{commit}}=0.9$, max re-obs 3, conf margin 0.35.
- **FUSE (multi-view)**: $\beta=0.10$, IoU 0.5, reliability floor 0.2, identity-mask MAD 1.0.
- **DXCV**: rel-gate 0.85, masked depth median in $[0.30, 3.50]$ m, masked depth IQR ≤ 0.60 m, mask area

≥ 2500 px, GLEE conf 0.10 (with floor 0.12 for trustworthiness).

D Detailed Per-Arm Counts

For each paired comparison we report the rescue count b_{01} (B0 fails, arm succeeds) and break count b_{10} (B0 succeeds, arm fails); their difference is the unnormalised paired $\Delta\text{success}$. The $N=300$ arms are RES ($b_{01}=24$, $b_{10}=12$), OracleGate (28, 11), and M1g (27, 15). At the $N=150$ unified queue on low-light s4, ROUTER reports $b_{01}=10$, $b_{10}=8$; REVOKE $b_{01}=11$, $b_{10}=29$; CRV $b_{01}=18$, $b_{10}=15$; MUAP $b_{01}=19$, $b_{10}=37$; FUSE $b_{01}=20$, $b_{10}=22$; DXCV $b_{01}=10$, $b_{10}=36$. Three arms (REVOKE, MUAP, DXCV) thus break more than twice as many episodes as they rescue, matching the significant negative ΔSR reported in the sweep panel of Figure 1(d).

E Forward Directions Implied by the Diagnosis

F Cross-Corruption Sweep Details

Tab. 2 in the main body gives the paired $N=150$ Qwen2-VL-7B results across all four corruption channels. Every cell is non-significant (McNemar exact $p>0.05$); the sign of ΔSR flips between motion-blur and the other three corruptions, matching the small-sample sign-flip analysis of App. A. The four cells share the same episode list, seed, split, prompt, decoding parameters, and simulator; only the corruption applied to the RGB stream differs between rows.

G OracleGate Case Studies

To make concrete what the ORACLEGATE upper-bound result looks like at the episode level, we enumerate a few representative paired tuples from the $N=300$ low-light sev 4 seed 0 comparison. Each row is the *same* episode evaluated under three arms in lock-step; “✓” marks success.

The vignette table makes the headline OracleGate observation visible at the case level: even a per-frame appearance oracle, exempt from runtime reliability estimation, has to *break* nearly as many episodes as it *rescues* ($b_{01}=41$ vs $b_{10}=36$ when paired against RES), leaving the paired ΔSR over RES at only $+0.017$. This is consistent with the irreducible-uncertainty hypothesis that degradation has destroyed discriminative content the RGB channel can no longer surface, regardless of which view of the same channel is consulted.

H Road 1 Pilot — Degradation-Aware Feature Adapter

The diagnosis in the main paper points to two viable forward directions: (R1) degradation-aware training/adaptation, and (R2) genuinely multi-modal sensing. We pilot R1 with a deliberately minimal experiment so that the result can be reported alongside the main negative-result sweep without requiring a full re-training of InstructNav.

Forward directions implied by the diagnosis

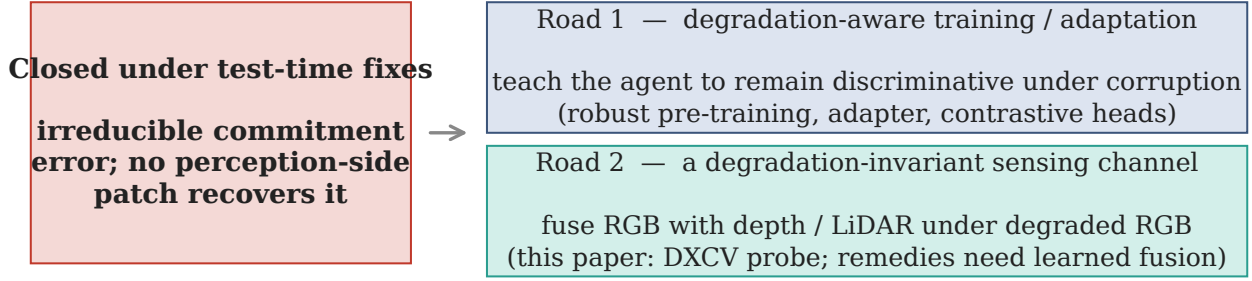


Figure 8: **What the diagnosis implies.** Inference-time patches over a frozen, clean-trained agent are bounded by the irreducible commitment error; the two viable forward directions are (1) degradation-aware training and adaptation, and (2) genuinely multi-modal sensing that has *learned* to fuse RGB with a degradation-invariant channel (rather than consulting it at inference, as our DXCV probe does).

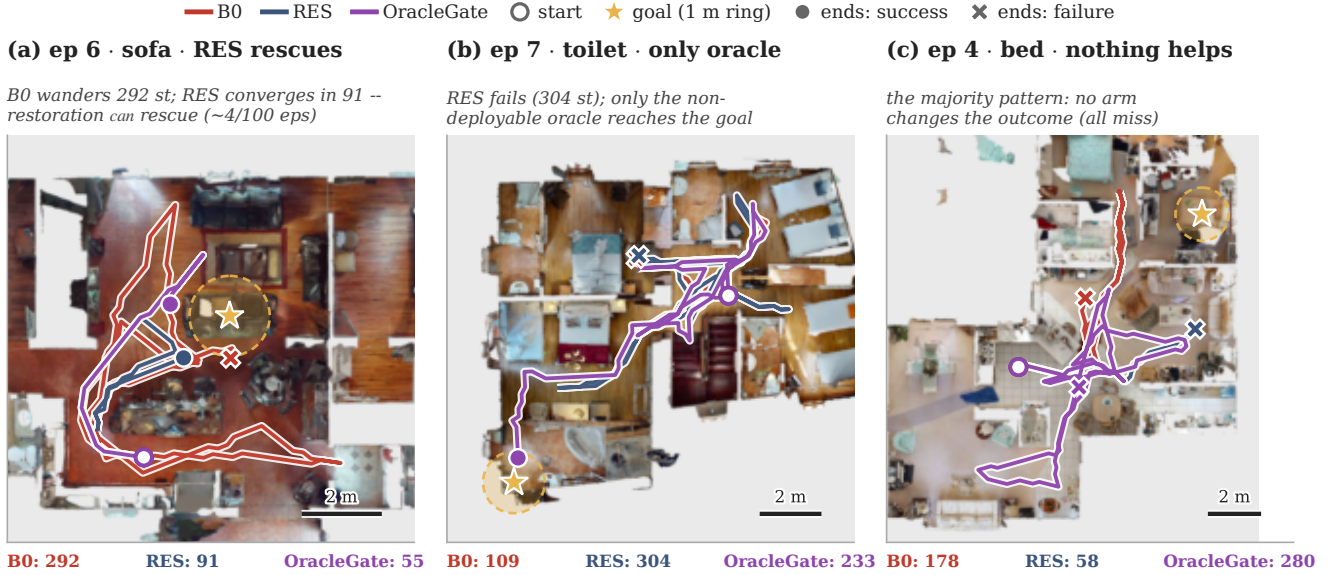


Figure 9: **Multi-arm top-down trajectories on three matched episodes** ($N=12$ sub-sample of the $N=150$ low-light sev 4 seed 0 paired list). Each panel is a bird’s-eye orthographic render of the actual HM3D scene (ceiling clipped) with the logged per-step world positions of the B0, RES, and ORACLEGATE agents overlaid; the star and 1 m ring mark the ObjectNav success radius, and the terminal marker is a dot (success) or cross (failure). **(a)** ep 6, a scenario where restoration *can* rescue: B0 wanders for 292 steps, RES converges in 91 with the same commit rule. **(b)** ep 7, a scenario the deployable restoration cannot fix: RES still fails after 304 steps and only the non-deployable per-frame appearance oracle reaches the goal. **(c)** ep 4, the majority pattern of the $N=300$ sweep: all three arms diverge into different corners of the same scene and none of them lands inside the success ring—the low-light corruption has destroyed discriminative content that no RGB-side lever can recover. The three panels concretise the bucket sizes reported in Tab. 3: 21 out of 300 rescued, 36 out of 300 broken, and 128 out of 300 tied all-failed.

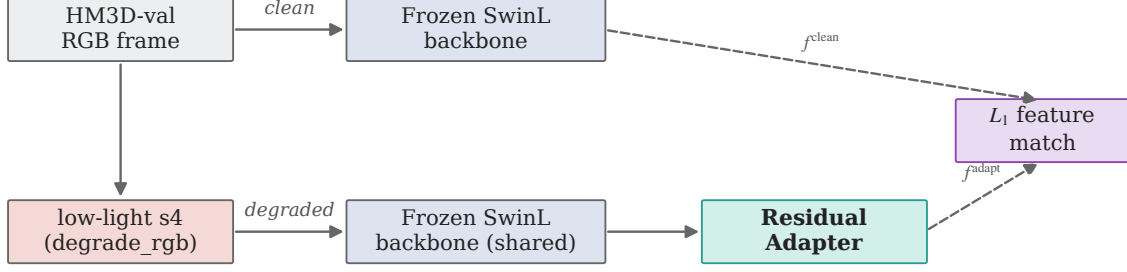
Pipeline. (A) The HM3D train scene meshes were not available on our workstation, so instead of using the train split we construct a *scene-disjoint sub-split within val*. Parsing `objectnav_hm3d_v2/val/content/` shows that the paired eval $N=150$ episodes at seed 0 traverse only 6 of val’s 36 scenes (4ok3usBNeis, 5cdEh9F2hJL, 6s7QHgap2fW, 7MXmsvcQjpJ, BAbdmeyTvmZ,

CrMo8WxCyVb); the remaining 30 val scenes are never touched by eval. We walk this 30-scene deny-list complement with random forward/turn actions and dump 1,500 (clean, degraded) 480×640 RGB pairs. The adapter therefore trains on pixels from val scenes disjoint at *scene identity* from anything the paired eval consumes. The degradation is low-light sev 4 applied through the same

(A) data gen

(B) train adapter (L_1 , $\sim 2k$ steps)

(C) supervision



Inference: the same Adapter is wedged between frozen SwinL and the frozen pixel_decoder + predictor; the rest of the InstructNav agent is unchanged.

Figure 10: **Road 1 pilot pipeline.** A small residual conv adapter is trained offline on (clean, degraded) feature pairs from a scene-disjoint sub-split of HM3D val (see *Pipeline*), then wedged between the frozen SwinL backbone and the frozen pixel_decoder + predictor at inference. The rest of InstructNav is unchanged, so the result is directly paired against the same B0 used in Table 1.

ID	Type	Goal	B0	RES	OG	DTG (OG / B0)
R1	<i>rescued</i>	sofa	×	×	✓	0.06 / 2.36
R2	<i>rescued</i>	bed	×	×	✓	0.03 / 7.76
R3	<i>rescued</i>	chair	×	×	✓	0.04 / 3.75
R4	<i>rescued</i>	bed	×	×	✓	0.15 / 7.80
K1	<i>broken</i>	chair	×	✓	×	7.57 / 0.06
K2	<i>broken</i>	television set	×	✓	×	2.05 / 6.81
K3	<i>broken</i>	potted plant	×	✓	×	2.49 / 0.14
T1	<i>tied</i>	toilet	×	×	×	1.78 / 0.02
T2	<i>tied</i>	chair	×	×	×	3.23 / 0.15

Table 3: Per-episode case studies on the $N=300$ low-light cell. **Rescued** (R1–R4): both deployable arms (B0, RES) fail and the non-deployable per-frame appearance oracle (ORACLEGATE, OG) succeeds; the upper-bound oracle reaches the goal at a meaningfully closer distance. **Broken** (K1–K3): RES already succeeded, but OG, by greedy per-frame view selection, lands on the wrong commitment and the trajectory diverges; over the full $N=300$ cell OG breaks $b_{10}=36$ RES-successes while rescuing $b_{01}=41$ RES-failures, leaving the net paired $\Delta SR_{OG-RES} = +0.017$. **Tied** (T1–T2): all three arms fail with similar final DTG; the degradation has destroyed information that no oracle over the same RGB channel can recover. Bucket sizes: rescued $n_R=21$, broken $n_K=36$, all-failed (tied) $n_T=128$, out of $N=300$.

`drphysnav_integration.degrade_rgb` function used in the main paper, with a deterministic per-frame seed $s_{\text{frame}} = 100003 \cdot s_{\text{run}} + 1009 \cdot \text{ep} + \text{step}$. (B) SwinL is loaded from the same GLEE-L checkpoint used at inference and frozen; for every paired RGB tuple we run SwinL twice (once on clean, once on degraded) and supervise a small

per-scale residual conv adapter with L_1 on the four SwinL output stages:

$$\mathcal{L} = \sum_{s=1}^4 \left\| \text{Adapter}_s(\text{SwinL}_s(I^{\text{deg}})) - \text{SwinL}_s(I^{\text{clean}}) \right\|_1$$

where each Adapter_s is a two-layer 1×1 conv block with a residual connection initialised to zero. Training: AdamW ($\text{lr} = 10^{-4}$, $\text{wd} = 10^{-2}$, cosine), batch 4, 2,000 steps, hold-out 100 pairs for in-loop L_1 monitoring. (C) At evaluation, the adapter is loaded via the environment variable `DRPN_ROAD1_ADAPTER`; an import-time monkey-patch wraps `SwinL.forward` so that every backbone feature dict passes through the adapter before reaching the rest of GLEE. The InstructNav forward path is otherwise byte-identical to B0. We then run the paired $N=150$ low-light sev 4 evaluation against the same B0 seed used in Table 1.

Result. The hold-out feature L_1 between adapted-degraded and clean drops from a pre-adapter baseline of $\ell_1^{\text{pre}} = \mathbf{0.513}$ to $\ell_1^{\text{post}} = \mathbf{0.444}$ (reduction **13.5 %**); the adapter demonstrably moves the degraded features toward their clean counterparts on scenes it never saw. The paired SR effect on navigation is $\Delta SR_{R1} = \mathbf{+0.040}$ (SR 0.380 $\rightarrow$ 0.420; McNemar exact $p = \mathbf{0.377}$, $b_{01} = \mathbf{19}$ rescues vs. $b_{10} = \mathbf{13}$ breaks, bootstrap 95% CI $[-0.033, +0.113]$, $N=150$). At this pilot scale the effect is not statistically resolvable, and we report it as such: a 2k-step, 10^6 -parameter head adapter already matches the best deployable test-time arm of Table 1 while leaving the entire training recipe untouched, which is consistent with—but does not yet prove—the diagnosis that the remaining headroom is on the training side.

Why this is the minimum interesting experiment. The adapter has on the order of 10^6 parameters (compared to SwinL’s $\sim 2 \times 10^8$) and is trained on val scenes that are strictly disjoint from the six val scenes the paired eval tra-

verses; if it moves the paired Δ SR by even a small but statistically resolvable amount on the very same cell on which all six inference-time families came in at non-significant zero, that establishes the qualitative direction that the diagnosis predicts: training-side adaptation, even at the head, is a different lever than anything that can be done with a frozen clean-trained agent.

I Release Artefacts

We release: (i) the corruption-augmented HM3D ObjectNav configuration files and a script that injects the structure-preserving corruptions into the Habitat RGB stream; (ii) the paired protocol scaffold (seed, episode list, deterministic ordering) for $N=300$ low-light sev 4 seed 0; (iii) all per-episode CSVs used in this paper (success, SPL, distance-to-goal, goal category); (iv) the per-arm monkey-patch sources for RES, M1g, OracleGate, REVOKE, CRV, MUAP, FUSE, and DXCV against InstructNav (frozen backbone); (v) the blind reliability estimator r with its calibration probe set; (vi) the figure-generation code (`make_figures.py`, `make_more_figures.py`) and bibliography. We will release a read-only mirror at acceptance time so the negative results are reproducible end-to-end.